

LLM Comparison Laboratory Activity

Exploring Content Generation Across Different AI Models

Submitted by:

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21st Century IT Skills 16-607

Ateneo de Davao University

SECTION 1: PROMPT

The following standardized prompt was used across both LLM platforms (ChatGPT and Claude):

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Act as an expert resume writer and career coach. Create a clean, modern, and highly professional one-page Microsoft Word resume based on the details provided below.

PERSONAL DETAILS:
Name: Josh Clarence R. Agulo
Email: agulo.joshclarence@gmail.com
Phone: 0997 227 5076 LinkedIn/
Linkedin: www.linkedin.com/in/joshagulo
Portfolio: https://github.com/JoshAgulo/portfolio

EDUCATION:
- BS INFORMATION TECHNOLOGY, Ateneo de Davao University, Graduating 2028 - Relevant Coursework:
Cybersecurity, Data Structures and Algorithms, Systems Information Architecture |

SKILLS:

Technical Skills: Cisco Networking, HTML, CSS, python, Java, Mysql, JavaScript, Reconnaissance OSINT
Soft Skills: The necessary soft skills for an IT professional

PROJECTS & EXPERIENCE:

School projects:

Talaan - Inventory Management System | Java 17, JavaFX, Maven Developed a full-featured POS and inventory management desktop application for sari-sari stores. Implemented custom data structures (Max Heap for top-selling items tracking, BST for temporal transaction search) and service-oriented architecture. Features include real-time inventory tracking, sales processing, supplier management, automated alerts, and comprehensive analytics dashboard.

TechCurrent News - Tech News Curation Platform | HTML5, CSS3, JavaScript Built responsive web application for personalized tech article curation featuring reading list management with LocalStorage persistence, dark/light mode toggle, and real-time UI updates. Implemented dynamic DOM manipulation, event-driven architecture, and modern CSS layouts (Flexbox/Grid) across 4 pages covering 6 tech categories. Demonstrated proficiency in vanilla JavaScript, semantic HTML, and responsive design principles.

Personal project

Automated Attack Surface Mapper that checks a company's domain for exposed infrastructure |Undergoing

ACHIEVEMENTS
(Optional): DOST UNDERGRADUATE MERIT SCHOLARSHIP, DEAN'S LISTER

FORMATTING & STYLE INSTRUCTIONS:
1. Ensure the layout is clean, minimalist, and easy to scan.
2. Use clear, bold section headings.
3. Keep the content concise and strictly limited to a one-page length.
4. Use strong action verbs and maintain a professional, confident tone suitable for an entry-level job application.
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SECTION 2: LLM OUTPUT

2.1 ChatGPT Output (LLM #1)

ChatGPT generated a comprehensive one-page resume with the following characteristics:

- Included a Professional Summary section highlighting key qualifications and career focus
- Integrated honors directly within the Education section for streamlined presentation
- Organized technical skills into four distinct categories with clear subcategorizations
- Provided detailed, action-oriented bullet points for each project with specific technical implementations
- Listed soft skills in a concise, single-line format with bullet separators

Josh Clarence R. Agulo

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LinkedIn: www.linkedin.com/in/joshagulo

GitHub: <https://github.com/JoshAgulo/portfolio>

PROFESSIONAL SUMMARY

Motivated Information Technology student with strong foundations in cybersecurity, networking, and software development. Experienced in building full-stack applications and implementing efficient data structures. Passionate about cybersecurity, automation, and developing secure, scalable systems.

EDUCATION

Bachelor of Science in Information Technology

Ateneo de Davao University — Expected Graduation: 2028

Relevant Coursework: Cybersecurity, Data Structures and Algorithms, Systems Information Architecture

Honors: DOST Undergraduate Merit Scholarship, Dean's Lister

TECHNICAL SKILLS

Programming & Development: Java, Python, JavaScript, HTML5, CSS3

Networking & Security: Cisco Networking, OSINT Reconnaissance, Network Fundamentals

Database & Tools: MySQL, Maven, Git, JavaFX

Core Competencies: Data Structures Implementation, Responsive Web Design, Debugging, Software Testing

PROJECTS

Talaan – Inventory Management System | Java 17, JavaFX, Maven

- Developed a POS and inventory management desktop application for sari-sari store operations.
- Implemented Max Heap for tracking top-selling items and Binary Search Tree (BST) for efficient transaction lookup.
- Designed real-time inventory tracking, sales processing, supplier management, and automated stock alerts.
- Built an analytics dashboard for monitoring inventory trends and sales performance.

TechCurrent News – Tech News Curation Platform | HTML5, CSS3, JavaScript

- Built a responsive web application featuring personalized article curation across 6 technology categories.
- Implemented LocalStorage persistence for reading list management.
- Developed dark/light mode toggle and real-time UI updates using event-driven JavaScript.
- Applied Flexbox and CSS Grid to create modern, responsive layouts.

Automated Attack Surface Mapper (Ongoing) | Python, Networking, OSINT Tools

- Developing a domain reconnaissance tool that identifies exposed infrastructure and potential security risks.
- Integrating OSINT techniques to automate discovery of publicly accessible assets.
- Focused on improving network visibility and cybersecurity awareness for small organizations.

SOFT SKILLS

Analytical Thinking • Problem Solving • Attention to Detail • Time Management • Team Collaboration • Clear Technical Communication

2.2 Claude Output (LLM #2)

Claude (Anthropic) generated a clean, modern resume with these features:

- No professional summary, diving directly into education and experience
- Created a dedicated Achievements & Honors section at the bottom of the resume
- Organized technical skills into three main categories with bold category labels
- Labeled soft skills as 'Professional Skills' rather than 'Soft Skills'
- Used visual formatting with blue accent colors and underlined section headers
- Provided concise yet comprehensive project descriptions with technical depth

JOSH CLARENCE R. AGULO

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EDUCATION

Bachelor of Science in Information Technology

Ateneo de Davao University | Expected Graduation: 2028

Relevant Coursework: Cybersecurity, Data Structures and Algorithms, Systems Information Architecture

TECHNICAL SKILLS

Programming & Development: Java, Python, JavaScript, HTML5, CSS3, MySQL

Technologies & Tools: JavaFX, Maven, Git, LocalStorage API, Responsive Web Design

Networking & Security: Cisco Networking, OSINT Reconnaissance, Security Infrastructure Mapping

PROFESSIONAL SKILLS

Problem-Solving & Analytical Thinking • Team Collaboration & Communication • Time Management • Adaptability & Continuous Learning

PROJECTS & TECHNICAL EXPERIENCE

Talaan - Inventory Management System | Java 17, JavaFX, Maven

- Developed full-featured POS and inventory management desktop application for sari-sari stores with real-time tracking and analytics dashboard
- Implemented advanced data structures (Max Heap for top-selling items, BST for temporal transaction search) and service-oriented architecture with supplier management and automated alert systems

TechCurrent News - Tech News Curation Platform | HTML5, CSS3, JavaScript

- Built responsive web application featuring personalized tech article curation with reading list management, LocalStorage persistence, and dark/light mode toggle
- Demonstrated proficiency in vanilla JavaScript, dynamic DOM manipulation, event-driven architecture, and modern CSS layouts (Flexbox/Grid) across 4 pages covering 6 tech categories

Automated Attack Surface Mapper | In Development

- Developing automated reconnaissance tool to map exposed infrastructure and security vulnerabilities across company domains using OSINT techniques

ACHIEVEMENTS & HONORS

- DOST Undergraduate Merit Scholarship - Department of Science and Technology
- Dean's List - Ateneo de Davao University

SECTION 3: COMPARISON TABLE

The following table provides a detailed comparison of both LLM outputs across key evaluation criteria:

Criteria	ChatGPT (LLM #1)	Claude (LLM #2)
Structure	Includes Professional Summary. Honors integrated within Education section. Four technical skill categories. Detailed bullet points for projects.	No professional summary. Separate Achievements section. Three technical skill categories with bold labels. Clean section headers with visual accents.
Clarity	Very clear information hierarchy. Professional summary provides immediate context. Comprehensive skill categorization makes technical abilities easy to identify.	Clean visual hierarchy with color accents. Direct approach without summary allows quick access to qualifications. Bold category labels enhance scannability.
Formatting	Traditional professional format. Clean black text on white background. Standard bullet points. Well-organized sections with consistent spacing.	Modern design with blue accent colors. Section headers have visual underlines. Professional color scheme enhances visual appeal while maintaining ATS compatibility.
Overall Quality	Excellent - Comprehensive and professional. Strong use of professional summary provides context. Detailed project descriptions showcase technical depth. Traditional format appeals to conservative industries.	Excellent - Modern and visually appealing. Clean design with strategic use of color. Efficient use of space. Strong visual hierarchy makes information easy to scan. Appeals to tech-forward companies.

SECTION 4: REFLECTION

Both ChatGPT and Claude produced high-quality, professional resumes, but each took a different approach. ChatGPT performed slightly better overall because it included a professional summary that immediately establishes my value proposition and career focus. This addition helps recruiters quickly understand my background without reading the entire resume. The integration of honors within the education section also created a more streamlined structure.

The most noticeable difference between the two outputs was their structural philosophy. ChatGPT favored a content-rich approach with detailed categorization and a professional summary, while Claude opted for a cleaner, more visual design with color accents and strategic formatting. ChatGPT organized technical skills into four categories versus Claude's three, and their treatment of achievements differed, ChatGPT embedded them in education while Claude created a separate section. Claude also renamed 'Soft Skills' to 'Professional Skills,' which sounds more mature and workplace-oriented.

For the final version, I combined the best elements from both. I kept ChatGPT's professional summary because it provides valuable context upfront, but I adopted Claude's modern visual design with blue accents and underlined section headers to make the resume more memorable. I also used Claude's 'Professional Skills' terminology and maintained ChatGPT's comprehensive technical skill categorization for completeness. The Attack Surface Mapper project benefited from both approaches, ChatGPT's inclusion of specific tools (Python, OSINT) combined with Claude's cleaner status indicator.

This exercise taught me that effective prompting requires clear, detailed instructions and that different LLMs interpret the same prompt through different lenses. ChatGPT leaned toward comprehensive content coverage, while Claude prioritized visual design and user experience. The key learning is that when working with AI tools, you should evaluate multiple outputs and synthesize the best elements rather than accepting a single result. Both models correctly interpreted my request for a one-page resume with clean formatting and professional tone, but their execution strategies differed in valuable ways. Understanding these differences helps me craft better prompts and make more informed decisions about which AI tool to use for specific tasks.

SECTION 5: FINAL REFINED RESUME

The following page contains the final refined resume that combines the best elements from both LLM outputs.

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PROJECTS

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- Implemented advanced data structures (Max Heap for top-selling items tracking, BST for efficient temporal transaction search) and service-oriented architecture
- Designed features including sales processing, supplier management, automated stock alerts, and inventory trend monitoring

TechCurrent News - Tech News Curation Platform | HTML5, CSS3, JavaScript

- Built responsive web application featuring personalized article curation across 6 technology categories with reading list management and LocalStorage persistence
- Implemented dark/light mode toggle, real-time UI updates using event-driven JavaScript, and modern responsive layouts with Flexbox and CSS Grid

Automated Attack Surface Mapper | Python, OSINT Tools (In Development)

- Developing automated reconnaissance tool to identify exposed infrastructure and potential security vulnerabilities across company domains
- Integrating OSINT techniques to automate discovery of publicly accessible assets and improve network visibility for organizations

PROFESSIONAL SKILLS

Problem-Solving & Analytical Thinking • Team Collaboration • Clear Technical Communication • Time Management • Attention to Detail • Adaptability & Continuous Learning